

Automated Priority Judgment for Sabre Fencing via Multimodal Pose Analysis

CS207 Final Report

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Abstract

Sabre fencing priority judgment requires referees to resolve sub-second kinematic decisions under the FIE right-of-way convention—a task prone to inconsistency at elite competition speed. We present an automated sabre referee system that combines YOLOv8-Pose keypoint estimation, audio transcription, and scoring-light detection to delimit competitive action windows, then extracts timestamped attack, lunge, and pause events to drive a cascading priority-judgment algorithm faithful to FIE rules [1]. The system correctly classified 9 of 11 test phrases, demonstrating the feasibility of pose-based priority adjudication in broadcast footage.

1 Introduction

In sabre fencing, a valid touch is awarded not merely to whichever athlete lands a hit, but to whichever athlete held *priority* at the moment of contact—a convention governed by the FIE right-of-way rules (t.87–t.106) [1]. When both fencers register touches within the same phrase, the referee must determine which athlete initiated the attack through precise temporal ordering of arm extension, forward commitment, and blade interaction. At elite speed, exchanges unfold in a few hundred milliseconds, making reliable human adjudication difficult and motivating the search for automated assistance [1].

Prior work has addressed related subproblems. Malawski and Kwolek [2] showed that wrist-mounted accelerometers can classify fundamental footwork categories. Vision-based approaches have since moved toward richer representations: Sunal *et al.* [3] demonstrated real-time move classification at touch time from monocular video, while Zhu *et al.* [4] introduced FenceNet for fine-grained footwork recognition. At the system level, Hanai *et al.* [5] deployed multi-camera blade tracking at the Tokyo 2020 Olympic Games, and Li *et al.* [6] combined skeleton-based action recognition with audio to achieve 89.1% scoring-side accuracy on 4000 clips. The most complete automated referee to date, FERA [7], targets foil fencing and reports 77.7% priority-judgment ac-

curacy, but acknowledges that phrase localisation remains a key bottleneck.

The present work targets sabre, whose cutting-action mechanics and priority structure differ fundamentally from foil. Our principal contribution is a principled, multimodal *temporal segmentation* mechanism—combining pose-based stillness detection, Whisper-based audio-cue identification [8], and scoring-light detection—that robustly delimits the competitive action window. Within this window, four kinematic events (baseline, attack onset, lunge onset, pause) are extracted and fed to a three-step cascading priority algorithm that formalises FIE Rules t.87–t.95.

2 Preliminary

2.1 Priority and Double-Hit Adjudication in Sabre Fencing

In fencing, when both competitors hit each other within a few milliseconds, both sides of the scoreboard will light up. [9] Then the referee has to make a decision on who’s point it is. In sabre, hit adjudication is governed by the convention of *priority* (right of attack), which determines which fencer is entitled to score when both athletes register touches within the same fencing phrase. According to the FIE Technical Rules, “the Referee alone decides as to the validity or the priority of the hit” (t.100) [1]. This makes sabre refereeing not merely a hit-detection task, but a complex process of temporal and tactical interpretation.

At elite competition speed, sabre exchanges often occur within a few hundred milliseconds, making reliable human adjudication difficult. In particular, *double-hit* situations represent one of the most challenging aspects of refereeing. The FIE rules explicitly distinguish between two cases:

“When during a phrase both fencers are hit simultaneously there is either a simultaneous action or a double hit” (t.106.1). [1]

A *simultaneous action* refers to the “simultaneous conception and execution of an attack by both fencers,” in which case “the hits exchanged are annulled for both fencers”

(t.106.1) [1]. By contrast, a *double hit* (*coup double*) is defined as “the result of a clearly faulty action on the part of one of the fencers” (t.106.2) [1]. Therefore, the referee must determine which fencer violated the tactical conventions governing priority.

The determination of priority is primarily based on the definition of a correctly executed attack. The rules specify:

“The attack is correctly carried out when the extending of the arm, the point threatening the valid target, precedes the initiation of the lunge or the *flèche*” (t.101.3) [1].

Furthermore, the rules state that “every attack ... must be parried or completely avoided” (t.101.1) [1]. Consequently, if one fencer establishes priority through a valid attack, the opponent must either perform a successful parry or evade the attack completely before launching a counter-action.

The strictness of sabre adjudication becomes especially evident in the regulations concerning double hits. Article t.106 enumerates multiple scenarios in which one fencer is considered solely at fault. For example, the defender loses priority “if he makes a stop hit on his opponent’s simple attack” (t.106.3.a) [1], or “if, instead of parrying, he attempts to avoid the hit and does not succeed” (t.106.3.b) [1]. Similarly, delayed defensive responses are penalized: after a successful parry, the riposte must be executed “immediately, without any hesitation or delay” (t.105.1) [1]; otherwise the attacker may regain priority through a remise or reprise (t.106.3.c) [1].

Conversely, the attacking fencer may also lose priority under specific conditions. The rules state that the attacker is solely counted as hit “if he initiated his attack when his opponent had his point ‘in line’, without deflecting the opponent’s weapon” (t.106.4.a) [1]. In addition, during compound attacks, the attacker loses priority “if, during a compound attack, he bends his arm or makes a momentary pause” (t.106.4.d) [1], or “if ... he is stop-hit one period of fencing time before he makes his final movement” (t.106.4.e) [1]. These rules require referees to evaluate subtle temporal relationships between blade motion, arm extension, and footwork in real time.

The FIE rules themselves acknowledge the inherent difficulty of these judgments:

“When there is a double hit, and if the Referee is unable clearly to judge from which side the fault has come, he must replace the competitors on guard” (t.106.5) [1].

This limitation motivates the use of computer vision techniques for sabre refereeing assistance. Since double-hit adjudication depends heavily on high-speed spatiotemporal analysis, automated systems capable of tracking arm extension, blade interaction, attack timing, and fencing phrases may substantially improve the consistency and interpretability of officiating decisions.

3 Related Work

Research relevant to this work spans three interconnected threads: sensor-based and early computer-vision approaches to fencing movement analysis; deep-learning methods for fencing action and footwork recognition; and integrated systems that approach automated scoring or priority judgment.

Sensor-Based and Early Computer-Vision Approaches

The earliest systematic attempts to quantify fencing movement relied on wearable hardware rather than video. Malawski and Kwolek [2] demonstrated that a single wrist-mounted accelerometer could classify six fundamental footwork classes across ten athletes, establishing the FFD benchmark that has since served as a reference baseline. Building on this, the same authors deployed a Kinect depth sensor for real-time footwork segmentation and lunge-parameter extraction, introducing a model-adaptive filtering algorithm to isolate footwork events from continuous velocity signals and an SVM to suppress false positives [10]. Although these systems established the feasibility of quantitative movement analysis in competition fencing, their dependence on contact hardware or dedicated depth cameras limits applicability in broadcast or tournament settings. On the computer-vision side, Athow [11] conducted one of the first systematic explorations of traditional CV to assist sabre, foil, and épée scoring judgment, laying the conceptual groundwork for later deep-learning approaches.

Deep Learning for Fencing Action Recognition

The transition to deep learning substantially advanced the state of automated fencing analysis. Sunal *et al.* [3] presented the first end-to-end deep-learning system for detecting the *touch-time* moment during live bouts, combining YOLOv3 for athlete detection with a ResNet-34 classifier to achieve 83% accuracy on the scoring instant. Zhu *et al.* [4] introduced FenceNet, a skeleton-based temporal convolutional network operating on 2-D pose keypoints, achieving 85.4% accuracy on the FFD footwork dataset; a bidirectional variant (BiFenceNet) raised this to 87.6%, demonstrating that directional motion context carries discriminative value for step classification. More recently, Ma *et al.* [12] applied YOLOv8x-Pose specifically to the sabre weapon, pairing keypoint features with a One-vs-Rest logistic classifier on a self-collected 2040-image dataset to improve multi-class movement recognition under high-speed conditions. At the cross-sport level, Lai *et al.* [13] proposed FACTS, a VideoMAE-based Transformer that classifies eight fencing action categories directly from raw video frames without explicit pose estimation, reporting 90% accuracy on a 13495-clip dataset; however, its end-to-end design does not expose the per-frame kinematic trajectories required for rule-based priority judgment.

Visual Tracking and Apparatus-Level Systems

Several systems have targeted the lower-level visual tracking problems specific to fencing. Hanai *et al.* [5] developed a real-time blade tracking and visualisation system deployed at the Tokyo 2020 Olympic Games, reconstructing 3-D blade-tip position from multi-camera feeds using YOLOv3 and CG-augmented training data to handle the severe motion blur of high-speed sabre blades. Quinn and Corcoran [14] proposed a more general framework integrating detection, tracking, and pose estimation into a single pipeline for combat sports, demonstrating automated performance-statistic generation from broadcast footage without sport-specific tuning.

Towards Automated Priority Judgment

The most direct precursors to the present work are systems that attempt to replicate referee judgment rather than merely classify isolated movements. Mo [15] proposed a multimodal AI referee combining a skeleton-based temporal convolutional network for action classification with an audio module that listens for blade-contact sounds, reporting 89.1% scoring-side accuracy across 4000 clips of international competition. Their audio-visual fusion strategy is directly relevant to the temporal segmentation mechanism proposed here, where acoustic signals (the referee’s *Allez* command and the scoring buzzer) serve as secondary indicators of analysis-window boundaries. The most complete automated referee prototype to date is FERA, proposed by Chen and Wang [7], which targets foil fencing: a 101-dimensional kinematic feature vector is constructed from 2-D keypoints, a Transformer multi-label classifier recognises footwork and bladework categories, and a distilled language model reasons over FIE priority rules, achieving 77.7% priority-judgment accuracy across 969 phrase clips. The authors acknowledge a substantial gap to practical deployment, attributing it in part to the difficulty of precisely localising the competitive action phrase within longer sequences—a challenge our analysis-window segmentation mechanism directly addresses.

The present work targets the *sabre* weapon, whose cutting-action mechanics and priority rule structure (FIE t.96–t.106) differ fundamentally from foil. Our principal contribution relative to prior work is a principled, multimodal temporal segmentation mechanism—combining pose-based stillness detection, audio-cue identification, and scoring-light detection—that robustly delimits the competitive action window, a component treated only superficially in all preceding systems [3, 7].

4 Core Design

4.1 Temporal Segmentation of the Competitive Action Window

A typical sabre competition video contains extended non-competitive periods interspersed with brief competitive phrases. Applying full analysis across the entire recording would impose unnecessary computational overhead and risk generating spurious events from warm-up movements that are structurally indistinguishable from genuine attack initiations. We therefore define a bounded *analysis window* $[T_{\text{open}}, T_{\text{close}}]$ and restrict all motion analysis, event detection, and priority judgment exclusively to frames within this interval.

4.1.1 Multimodal Event Monitoring

Window boundaries are inferred from four complementary signals spanning two modalities:

- **Audio:** referee vocal commands (*Allez*) and electronic scoring buzzer activations.
- **Vision:** scoring apparatus indicator lights and athlete pose states (stillness and post-hit static posture).

Window open: pose-based stillness (primary). The window opens when both athletes hold a stable on-guard position. Joint angles (elbow, front knee, trunk lean) are derived from YOLOv8x-Pose [8]; a baseline calibration event fires when the standard deviation of every tracked angle remains below $\sigma_{\text{max}} = 4.0^\circ$ over $N_{\text{still}} = 10$ consecutive frames (≈ 333 ms at 30 fps). The resulting per-athlete angle means serve as baseline references for all subsequent delta-based thresholds, making the system robust to inter-athlete variation in guard posture.

Window open: audio indicators (secondary). The referee’s *Allez* command is detected via faster-Whisper [16] applied to the FFmpeg-extracted audio track, and buzzer onset is located by a spectral energy detector. Audio signals are treated as fallback evidence because microphone placement and crowd noise vary across venues, causing occasional missed or mis-transcribed detections [6, 15].

Window close: scoring light activation. The window closes upon confirmed bilateral activation of the scoring apparatus lights, detected in HSV colour space on the lower 35 % of each frame. A light-on event requires a colour blob exceeding $A_{\text{min}} = 5000 \text{ px}^2$ persisting for at least two consecutive frames to suppress false activations from reflections or LED flicker. The window is extended by $\Delta t_{\text{close}} = 10$ frames beyond the first confirmed event to preserve any ongoing movement trajectory at the moment of touch registration [1]. All detections outside $[T_{\text{open}}, T_{\text{close}}]$ are discarded and excluded from the priority judgment.

4.2 Motion Detection

Determining priority in sabre fencing requires resolving the precise temporal ordering of each athlete’s offensive actions. Raw pose estimator output is insufficient for this purpose; structured motion events with frame-level timestamps must first be extracted.

This extraction is complicated by motion blur, which is prevalent during fast limb movements such as attack extensions and lunges in standard 30 fps broadcast footage, causing unstable keypoint estimates during precisely the phases that matter most for event timing. Inspired by Bright et al. [17], we apply a rolling mean over a 3-frame window (approximately 100 ms at 30 fps) to all joint angle measurements prior to event detection, reducing frame-to-frame noise without introducing perceptible temporal lag. Position-based fields used for velocity computation are excluded from smoothing to preserve kinematic accuracy.

We detect four classes of motion events within each analysis window. **Baseline calibration** establishes each athlete’s static on-guard joint angles as a personalised reference for all subsequent detectors. **Attack extension onset** marks the frame at which the sword arm begins a sustained straightening, identifying who initiated the offensive action and when. **Lunge detection** confirms full offensive commitment, distinguishing a genuine attack from a feint or preparatory step. **Pause detection** identifies interruptions in forward momentum after launch, which cause the athlete to forfeit priority under the right-of-way rules. These four events provide the timestamped evidence on which the priority judgment algorithm operates.

4.2.1 Baseline Detection

The baseline captures each athlete’s static on-guard joint angles before the action begins, serving as the reference zero-point for all subsequent motion thresholds.

Each frame, the current angle dict is pushed into a sliding window W of fixed length. Stability is evaluated once the window is full. Let $\{\theta_i^e\}$ and $\{\theta_i^k\}$ denote the elbow and front-knee angle sequences over the window. The window is declared **stable** if all five conditions hold simultaneously:

$$|W| \geq N_{\min}, \quad N_{\min} = 10 \quad (1)$$

$$\bar{\theta}^e \in [\theta_{\min}^e, \theta_{\max}^e] = [50, 160] \quad (2)$$

$$\bar{\theta}^k \in [\theta_{\min}^k, \theta_{\max}^k] = [80, 165] \quad (3)$$

$$\sigma(\theta^e) < \sigma_{\max} = 4.0 \quad (4)$$

$$\sigma(\theta^k) < \sigma_{\max} = 4.0 \quad (5)$$

On capture, the baseline $\mathcal{B}$ stores the window mean of the following quantities:

- Elbow angle $\bar{\theta}^e$
- Shoulder angle $\bar{\theta}^s$
- Front knee angle $\bar{\theta}_f^k$ and rear knee angle $\bar{\theta}_r^k$

- Trunk lean $\bar{\alpha}$
- Stance width $\bar{d}_{\text{stance}}$ (px)
- Centre-of-mass x-position $\bar{x}_{\text{CoM}}$
- Wrist height $\bar{y}_w$

Shared capture. If side A stabilises while side B has not, B ’s baseline is immediately computed from whatever frames B has accumulated at that moment, avoiding an indefinite wait.

4.2.2 Lunge Detection

In sabre fencing, the lunge is the primary offensive movement and the clearest physical expression of an attack. It is executed by driving the front leg forward while simultaneously extending the sword arm, allowing the fencer to cover distance rapidly and deliver a hit. Because the lunge commits the athlete’s full bodyweight in one direction, it is largely irreversible once initiated — making its precise onset a critical timestamp for priority judgment.

Lunge detection is based on three biomechanical conditions that must be satisfied simultaneously within the active analysis window. For each athlete, the system continuously computes three metrics from the YOLOv8-Pose keypoints:

A lunge onset event is triggered when all three conditions are met:

- The front knee angle exceeds 150° , indicating full forward leg extension.
- The inter-thigh angle exceeds a side-specific threshold — 100° for the left athlete and 120° for the right — to confirm leg separation.
- The stance width ratio exceeds $2.0\times$ the baseline, confirming that the step length has grown well beyond the normal on-guard stance.

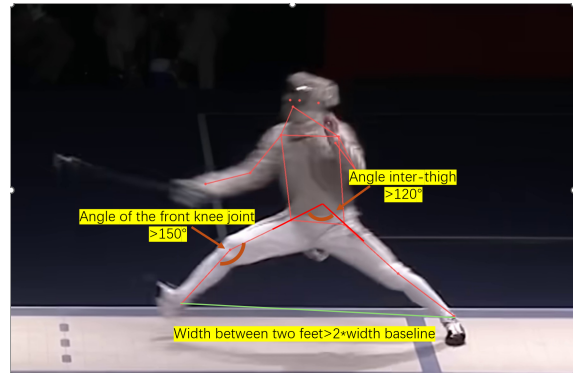


Figure 1: Lunge detection

The side-specific thigh threshold accounts for the projection difference between two athletes facing opposite directions in a single side-view camera, as discussed in the preceding section. These threshold are set after testing all the current test case and output.

4.2.3 Attack Detection

Attack initiation is identified by two absolute joint angle conditions that must be satisfied simultaneously for a minimum of two consecutive frames.

- **Elbow angle** bigger than 140° : the interior angle at the right elbow, formed by the vectors toward the right shoulder and right wrist, indicating that the sword arm has reached a substantially extended position.
- **Shoulder angle** bigger than 60° : the interior angle at the right shoulder, formed by the vectors toward the right hip and right elbow, confirming that the arm has been raised and projected forward away from the body.

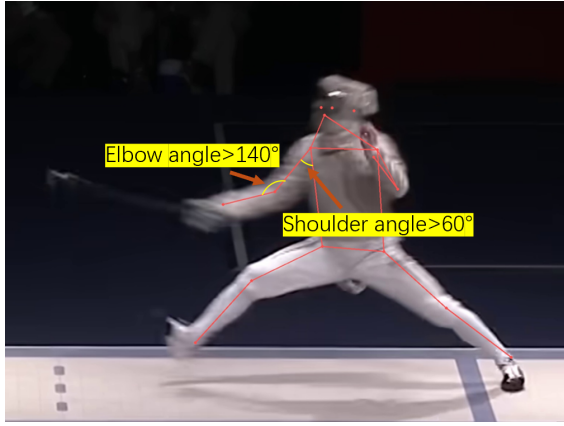


Figure 2: Attack detection

The two-frame confirmation window serves as a debounce mechanism: a single frame in which both thresholds are met is not sufficient to register an attack, preventing transient keypoint noise or motion-blur artifacts from generating false positives. Once confirmed, the event is recorded as **ATTACK-START** with its frame timestamp, and no further attack events are emitted for that athlete within the same analysis window.

Both angles are measured on the anatomical right arm for both athletes, as sabre fencing convention requires the sword to be held in the right hand regardless of which side of the screen the athlete occupies. Prior to detection, all joint angles undergo a three-frame rolling average to reduce noise introduced by motion blur during fast limb movements, following the approach of Bright et al. [17].

4.2.4 Pause Detection

A pause event signals that an athlete’s forward action has been interrupted after launch, causing loss of priority. Let v_f denote front-ankle speed and v_{CoM} the signed centre-of-mass velocity, both in px/frame.

The detection gate requires a **LAUNCH** event to have already been emitted for the same athlete. Each subsequent frame is tested against the **stopped** predicate:

$$\text{stopped} \iff v_f < \tau_f \wedge |v_{CoM}| < \tau_c, \quad \tau_f = \tau_c = 2 \text{ px/frame} \quad (6)$$

Let n be the number of consecutive frames satisfying *stopped*. A single **ATTACK-PAUSE** event is emitted the first time:

$$n \geq N_{\text{pause}}, \quad N_{\text{pause}} = 4 \text{ frames} \quad (7)$$

If any frame breaks the predicate, n resets to zero. The event fires at most once per athlete per action window. The resulting flags are:

- **pause_L / pause_R** — set to **True** on **ATTACK-PAUSE** emission
- **step.back_L / step.back_R** — set to **True** if an **L_STEP_BACK** or **R_STEP_BACK** event occurs at any frame $t \geq t_{\text{launch}}$

Either flag being **True** causes the athlete to forfeit priority in the judgment algorithm. The dual-signal requirement on v_f and v_{CoM} prevents false positives from isolated arm swings, which raise v_{CoM} without halting the feet, or brief foot-plant phases where $v_f \approx 0$ but forward momentum continues.

4.3 Priority Decision Workflow

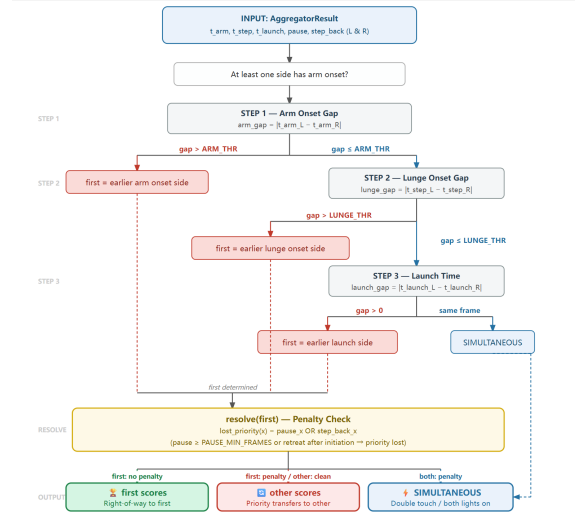


Figure 3: Sabre priority workflow

In sabre fencing, the assignment of a valid touch is governed by the FIE right-of-way convention, which stipulates that when both fencers are hit simultaneously, the touch is awarded to whichever fencer held *priority* at the moment of contact. Priority is established by the initiative of attack: a fencer who extends their weapon arm and continuously threatens the valid target seizes priority, which the opponent can only reclaim by deflecting the blade or after the attacker’s action is completed [1, t.87–t.95]. If neither fencer

can be awarded priority — because both initiated their actions indistinguishably at the same moment — the result is declared **simultaneous**, and no point is awarded to either side.

The automated decision algorithm formalizes this convention as a three-step cascading priority judgment, illustrated in Figure 3. The system operates on timestamped kinematic events extracted per frame: arm-onset time (t_{arm}), lunge-onset time (t_{lunge}), launch time (t_{launch}), and two binary penalty flags indicating whether a fencer paused or retreated after their own launch.

Step 1 — Arm Onset. The algorithm first compares the arm-onset times of both fencers. In accordance with FIE Rule t.87, the fencer who extends first establishes the attack. If the gap between the two arm-onset frames exceeds a configurable simultaneity threshold $\tau_{\text{arm}} = 2$ frame, the earlier side is identified as the first mover and passed to the **resolve()** function. If the gap falls within the threshold — meaning both fencers raised their arms near-simultaneously — the algorithm cannot distinguish initiative from arm motion alone and proceeds to Step 2.

Step 2 — Lunge Onset. Step 2 compares lunge-onset times. A lunge represents the committed forward displacement of the body and corresponds to the FIE notion of a fencer “launching the attack with a continuous forward movement” [1, t.89]. If the lunge-onset gap exceeds $\tau_{\text{lunge}} = 2$ frame, the earlier side is treated as the initiating fencer. Otherwise, the algorithm falls through to Step 3.

Step 3 — Launch Time. Step 3 compares raw launch times with 10 frame simultaneity threshold: If the fencers’ launch time is bigger than 10 frames, then there is a priority. If not, it is considered as a simultaneous.

Forfeiture Check (resolve()). At every step where a first mover is identified, the verdict passes through the shared **resolve()** function, which implements the FIE forfeiture rule [1, t.91]: a fencer who pauses in place or retreats after initiating their own attack is deemed to have relinquished priority. Formally, let $p_i \in \{0, 1\}$ denote the penalty flag of fencer i , where

$$p_i = \mathbf{1}[\text{pause}_i] \vee \mathbf{1}[\text{step_back}_i]. \quad (8)$$

Given the identified first mover f and the opposing side $\bar{f}$, the verdict is determined as follows:

$$\text{verdict} = \begin{cases} f \text{ scores} & \text{if } p_f = 0, \\ \bar{f} \text{ scores} & \text{if } p_f = 1 \text{ and } p_{\bar{f}} = 0, \\ \text{SIMULTANEOUS} & \text{if } p_f = 1 \text{ and } p_{\bar{f}} = 1. \end{cases} \quad (9)$$

A SIMULTANEOUS verdict corresponds to a double hit, in which both scoring lights illuminate and no point is awarded.

The same outcome is returned if the cascade exhausts all three steps without resolving a clear first mover.

This cascading structure ensures that priority is assigned at the earliest and most decisive kinematic signal available, while the forfeiture check at each stage preserves fidelity to the FIE principle that an interrupted or retreating attack cannot claim the touch.

5 Result

5.1 Joint angle timeline

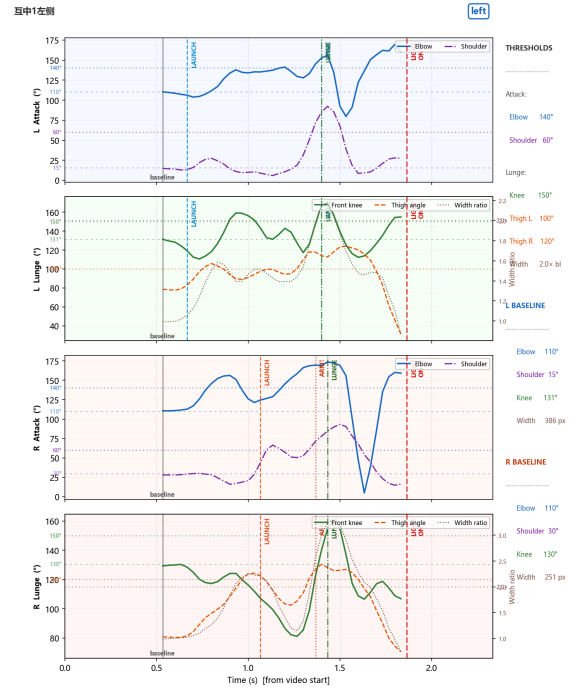


Figure 4: Joint angle timeline

The four subplots shows Left Attack, Left Lunge, Right Attack and Right Lunge. It track elbow angle, shoulder angle, front knee angle, inter-thigh angle, and stance width ratio for both fencers across the action window (0.5s–2.0s of video time). The grey vertical line marks the baseline capture moment — the on-guard posture before either fencer begins moving.

For the left fencer’s attack (L Attack subplot), the elbow angle begins near the baseline of 110° and climbs steadily, crossing the 140° detection threshold and peaking beyond 175° around the 1.4s mark — a clear, full arm extension. The shoulder angle rises from a baseline of 15° to above 90° , confirming the sword arm is being driven forward rather than simply straightening at the elbow alone. The LAUNCH event is marked at 1.0s, and ARM↑ fires slightly after 1.4s. The left lunge (L Lunge) shows the front knee angle rising from 131° baseline and briefly exceeding the 150° threshold around the same time, with the inter-thigh angle and width

ratio both increasing in sync — consistent with a coordinated lunge.

For the right fencer, the pattern is similar in shape but delayed. The R Attack elbow also reaches 175° , but the LAUNCH event occurs roughly 0.4s later than the left fencer’s, and ARM $\uparrow$ is nearly coincident with the left fencer’s arm event at 1.4s. Notably, the right fencer’s shoulder baseline sits near $20\text{--}30^\circ$, suggesting a lower, more withdrawn guard position at the start. The R Lunge front knee crosses 150° at almost exactly the same moment as the left fencer’s lunge, making lunge timing inconclusive as a tiebreaker.

The LIGHT ON (red dashed line, 1.8s) marks when both scoring lights activated, closing the analysis window.

5.2 Event timeline

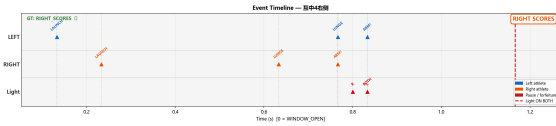


Figure 5: Event Timeline

The timeline shows each detected event as a marker on three lanes (LEFT, RIGHT, Light), with time relative to window open ($t = 0$).

The most decisive feature is the launch gap: the left fencer’s LAUNCH marker appears at 0.15s, while the right fencer’s LAUNCH does not appear until 0.5s — a difference of roughly 9–10 frames at 30fps, well above the 10-frame simultaneous threshold. This means the left fencer initiated their attacking movement significantly earlier.

By contrast, ARM and LUNGE events for both fencers cluster tightly around 0.85–0.95s, showing that once the action was underway, both fencers extended their arm and completed their lunge at nearly the same time. Without the earlier launch advantage, the outcome would have been called simultaneous at the arm and lunge steps.

The light lane shows the left scoring light (L) activating slightly before BOTH lights at 1.0s, consistent with the left fencer landing the touch first. The system correctly resolves the priority cascade at Step 3 (launch gap) and returns LEFT SCORES, matching the ground truth label.

6 Discussion

The system correctly classified 9 out of 11 test videos. One misclassification involved a *flying lunge*, an atypical technique where the fencer leaves the ground entirely rather than driving the front foot forward along the piste. Because the system’s lunge detector relies on front knee extension and stance width growth — both of which follow a different trajectory in a flying lunge — the event was either missed or mis-timed, leading to an incorrect verdict. This highlights a fundamental limitation: the detection logic is tuned to the

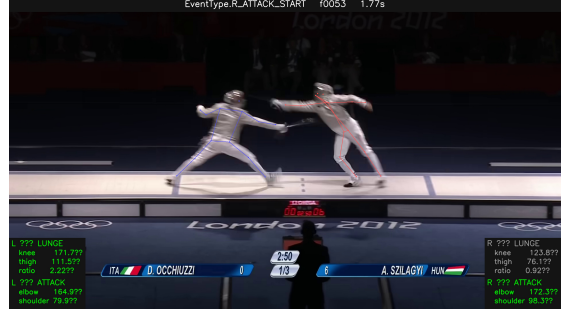


Figure 6: Caption

biomechanics of a standard lunge and does not generalize to less common attacking patterns. The system uses absolute joint angle thresholds (e.g. elbow ≥ 140 , front knee ≥ 150) rather than per-athlete deltas from baseline. Absolute thresholds are more robust in practice because they do not depend on the quality of baseline calibration — a noisy or early-captured baseline will not corrupt the detection. The trade-off is reduced generalisation: athletes with naturally more extended or more flexed on-guard postures may trigger thresholds too early or too late. A hybrid approach combining an absolute floor with a relative offset could address this in future work.

An asymmetry was observed between left and right athletes in the inter-thigh angle measurement. Because both fencers stand in a side-on stance relative to the camera, the apparent angle between the thighs depends on which direction each fencer is facing. The fencer facing left presents their legs at a different projected angle than the fencer facing right, even for identical physical postures. This introduced a systematic bias that required separate thresholds for each side ($LUNGE_THIGH_ANGLE_L = 100$, $LUNGE_THIGH_ANGLE_R = 120$). A 3D pose estimation approach or multi-view setup would eliminate this perspective-dependent error.

Two further limitations are worth noting. First, YOLOv8 keypoint estimates become noisy during the fastest phase of a lunge due to motion blur; temporal smoothing over a 3-frame window mitigates this but introduces a small lag. Second, all spatial measurements (stance width, CoM velocity) are in pixel units, meaning detection thresholds are sensitive to camera distance and zoom level. Calibrating against the 4-metre piste markings visible in competition footage would convert these to metric values and make the system camera-independent.

7 Conclusions

We have described an automated sabre referee system that segments competitive action windows from broadcast footage using multimodal cues and infers priority from pose-derived kinematic events under the FIE right-of-way convention [1]. Evaluated on 11 annotated phrases, the system achieves an accuracy of 9/11 ($\approx 82\%$).

Two misclassifications expose the principal limitations.

The lunge detector relies on absolute joint-angle thresholds calibrated to the standard lunge biomechanics and fails on atypical techniques such as the flying lunge. Additionally, all spatial measurements are pixel-based, making them sensitive to camera distance and zoom—a limitation that 3-D pose estimation or a calibrated multi-view setup would address, following the direction of Hanai *et al.* [5]. Future work should explore per-athlete relative thresholds, incorporate temporal models such as TCNs [12] for action segmentation, and extend evaluation to a larger, publicly available dataset.

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